Combining Multi-vision Embedding in Contextual Attention for Vietnamese Visual Question Answering

Anh Duc Nguyen[1,2], Tung Le (✉)[1,2], Huy Tien Nguyen[1,2]

[1] Faculty of Information Technology, University of Science, Ho Chi Minh, Vietnam
[2] Vietnam National University, Ho Chi Minh city, Vietnam
20C11016@student.hcmus.edu.vn
{lttung, ntienhuy}@fit.hcmus.edu.vn

Abstract. Visual question answering (VQA) is a challenging vision-and-language task that has recently gained high interest from many researchers. Existing methods adopt the strategy to extract object regional features, which effectively captures the local context of the image, but on the other hand, causes the loss of global context. Motivated by the strength of Vision Transformer, we propose to extract and combine both global and local contextual visual features for to be a Multi-vision Embedding for better image understanding. We also propose a Multimodal Contextual Attention method to adaptively learn the attentional embeddings of different contexts from image and question that are guided by each other. While most VQA models are optimized for the English dataset, few works are accomplished for the Vietnamese language. Hence, in this paper, we perform extensive experiments on a Vietnamese VQA dataset to evaluate and demonstrate the effectiveness of our approach. Our proposed model achieves an accuracy of 60.76% on the test set and outperforms former baseline methods in the Vietnamese language.

Keywords: Visual question answering · Attention mechanism · Vision Transformer · ResNet · BERT.

1 Introduction

Computer Vision (CV) and Natural Language Processing (NLP) have made significant breakthroughs in the last decade. Especially, the appearances of attention mechanism [1] are a milestones in both CV and NLP area. Together with the traditional neural networks such as Convolutional Neural Network (CNN) architecture in CV and Long Short-tem Memory in NLP, these attentive approaches improve the performance in most important tasks such as Image Classification [2], Object Detection [3], Machine Translation [4]. With the advancement in both fields, more researchers can effectively solve more complex problems that

(✉) Corresponding author: Tung Le – lttung@fit.hcmus.edu.vn

required the understanding of both vision and language, such as Visual Question Answering (VQA) [5, 6], Visual Captioning [7], and Visual Reasoning [8]. Compared to other vision-and-language tasks, VQA is a more challenging task that requires not only a deep understanding separately in image and question but also an efficient method to learn joint feature representations of them in a single framework to predict the correct answer.

Recent works have shown substantial improvement in the VQA task [8–10]. However, most VQA frameworks focus only on VQA in English, there is a lack of datasets for other languages, especially Vietnamese. Khanh et al. [11] introduced ViVQA, a Vietnamese VQA dataset translated from VQA dataset [12] and certain baseline methods but did not achieve reliable results. Motivated by this, we develop our model to enhance the performance of the VQA system in Vietnamese. We evaluate our model on the ViVQA dataset and can outperform the former baselines.

Most modern VQA models hold three major components: (i) image understanding for visual features, (ii) question embedding for textual features, and (iii) fusion of these features as joint feature representations.

For image understanding, early approaches [13, 14] used CNN to extract visual features as a feature map, then Anderson et al. [8] proposed bottom-up attention to extracting regional object-level features [15] and gained wide popularity in the VQA community. However, these methods have limitations, while CNN and bottom-up attention can effectively capture the local contextual visual features, they are ignoring the global contexts in an image. Vision Transformer (ViT) [16], a visual Transformer-based architecture [1] originally designed for text-based tasks, is proven to capture more long-range dependencies in visual extraction [17]. To overcome this limitation, we propose to take advantage of both CNN and ViT models to simultaneously attend to both global and local contextual visual features of the image. The combination of our image understanding can obtain more efficiency and performance than a single module.

For question embedding, various methods were applied to extract textual features such as LSTM [18], GloVe [19], etc. In our framework, we try to use the pre-trained BERT model [20], the state-of-the-art method in many NLP tasks, to enhance the question understanding in Vietnamese language.

Beyond the image and question understanding, the most important part of a VQA model is to establish a fine-grained features fusion to align the image and question semantically, which presents the key challenge. Early methods simply used linear fusion such as concatenation, summation, or element-wise product. Recent efforts have shown that the attention mechanism [1] is a powerful method for encoding the relation between visual and textual features. Since the VQA model aims to focus on the specific parts of the image that are relevant to the given question to find the correct answer, simulating the human-like reasoning process. Consequently, we design multi-branch attention that hierarchically fuses both global and local contextual visual features with textual features by leveraging the transformer encoder with top-down attention.

In summary, this paper makes the following contributions: (1) a method to simultaneously learn both global and local contextual visual features of the image; (2) a multi-branch contextual attention fusion that hierarchically integrates visual and textual features; (3) extensive experiments on a Vietnamese VQA dataset to demonstrate the effectiveness and outstanding performance of our model compared with the former baselines.

2 Related works

Visual understanding of VQA models. In VQA tasks, many proposed systems used image features extractor from a single stream: early methods tried to extract features from one or more layers of CNN network such as VGG [21], ResNet [22]; modern approaches used the bottom-up attention [8] which extracts object attended regional proposals from Faster-RCNN [15]. Moshiur et al [23] proposed to jointly attend both object-level and image-grid features to learn objects and scenes context. Recently Vision Transformer [16] and its variants [24, 25] open a promising era in Computer Vision, but they seem unpopular in vision-and-language tasks due to its weak inductive bias when training on small datasets. In other domains, many efforts like ConViT [26], Efficient ViT [27] try to combine both CNN network and Vision Transformer to harness both local and global information of an image and achieve competitive results.

Attention mechanism for joint features representation. In the early stage, many simple techniques were applied for multimodal fusion: concatenation, element-wise product [12]; or bilinear pooling methods [14]. Then attention mechanism became widely used and an important method for learning features fusion. Models [8, 13] with attention mechanism adaptively attend to meaningful areas in the image based on the question. Researchers then realized the importance of co-attention to both visual and textual information simultaneously such as: MCAN [9], DMBA [28], etc. To go beyond, certain efforts like ViLBERT [10], OSCAR [5] attempt to embed text and image together as an input sequence, and then feed into a Transformer-based architecture that automatically discovers relations between the two domains.

VQA datasets for non-English language. Recent advancement in VQA task is mostly built based on English datasets. The VQA dataset [12] and Visual Genome [29] are the two most popular datasets in English. Many researchers attempted to build their own datasets in their own languages by two methods [30]: (i) utilize sources like Wikipedia, newspaper, and human annotators to generate question-answer pairs; (ii) translate from existing English VQA datasets into their own language. In Vietnamese, recently, Nguyen et al. proposed ViMMRC [31], a multiple-choice dataset from school supplements of grade, and UIT-ViQuAD [32], a dataset which uses Wikipedia as a data source. Then Khanh et al. [11] proposed ViVQA, a Vietnamese dataset which is translated and adjusted from English VQA dataset [12].

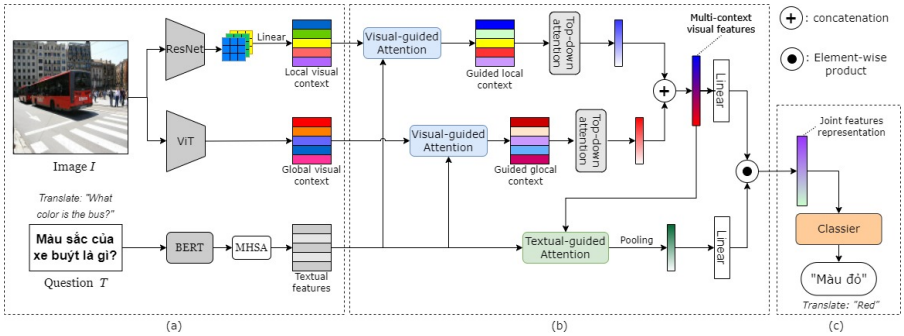


Fig. 1: Overview of our proposed Multi-vision Contextual Attention model

3 Methodology

In this section, we introduce our model architecture for the Vietnamese VQA task. Given an image I and a question T pair, the model will try to predict the best matching answer $\hat{a}$ from the set of candidate answers $\mathcal{A}$:

$$\hat{a} = \underset{a \in \mathcal{A}}{\operatorname{argmax}}\, p(a|I, T, \Theta) \tag{1}$$

where Θ denotes the set of parameters of the model.

The overview of our model is illustrated in Figure 1, the model consists of 3 main modules: (a) the features extraction networks to learn the image and question understanding; (b) the multi-branch contextual attention fusion to fuse visual and textual features; (c) the classification network to predict the answer.

3.1 Multi-vision embedding for image understanding

Visual features extraction plays a key role in the VQA task, grid features from CNN networks and object regional features from Faster-RCNN [15] are the two most popular techniques. However, these two models heavily depend on convolutional filters to embed visual attributes throughout their layers, where each layer convolves defined regions of the input and passes to the next layer, alias the local receptive fields. This such local connectivity can lead to the loss of global context of the image [33].

Vision transformer (ViT) [16], on the other hand, offers a multi-head self-attention mechanism to learn contextually the relevant information on the image. This mechanism is proven with the same effect as the local receptive field, however, is much more flexible. The flexibility of attention can capture more long-range dependencies information of the overall image, not just bounding to local regions. This results that global contextual information of ViT's receptive field is indeed larger than CNN in lower layers and can maintain that information through the network, while CNN receptive fields grow local to global. This

assurance is proven in the recent study of Raghu et al. [17], which established that ViT can strongly preserve the global spatial information even at the higher layers, [22], which helps gaining more global and attentive features.

Global and local contextual visual features can have a great impact on image understanding in the VQA task. While the local contextual features refer to the detailed meaningful regions such as objects, parts, and attributes; the global contextual features indicate the scene semantics or the interactions between objects of the entire image. The combination of these two features can help consolidate each other to achieve the best performance compared to single feature learning.

Therefore, for our model, we take advantage of both global and local visual information by extracting features from ViT and ResNet, respectively, in two separate branches. For ResNet, we obtain the local and residual contextual features $V_L \in \mathbb{R}^{M \times n_L}$, where M denotes the number of spatial grid locations of the extracted feature map with n_L dimensions. For ViT, we obtain the global and attentive contextual features $V_G \in \mathbb{R}^{N \times n_G}$, where N denotes the sequence length of the visual feature with n_G dimensions. Simplistically, given the input image I, we can formulate as:

$$V_L = ResNet(I) \tag{2}$$

$$V_G = ViT(I) \tag{3}$$

3.2 Question understanding

By considering the input question T as a sequence of words. We embed these words into textual feature vectors with 768-dimensional using BERT model. BERT [20], which stands for Bidirectional Encoder Representations from Transformers, is a state-of-the-art language model in many NLP tasks, the strength of BERT is its bidirectionality and attention to learn contextual embeddings. BERT divides the sentence into subwords, then maps to a set of embeddings E. Each embedding $e \in E$ is computed by summing: 1) a token embedding e_t, respective to the subword, 2) a segment embedding e_s, indicating which part of sentence the token comes from (in a pair of sentence) and 3) a position embedding e_p, indicating the position of the token in the sentence. The input embeddings E are then passed through a multi-layer Transformer that accumulates a contextual representation of the sentence. As we develop our model especially for Vietnamese, we utilise phoBERT [34], a variant of BERT which is pre-trained intensively on Vietnamese Wikipedia and news corpus, hence phoBERT is effcient for downstream Vietnamese NLP tasks. We can formulate as:

$$Q_T = phoBERT(T) \tag{4}$$

where $Q_T \in \mathbb{R}^{Sx768}$ is output features vector of the input question with length S. After obtaining the question features from phoBERT, we use a Multi-Head Self-attention (MHSA) to enhance the features further. Detail of the MHSA mechanism will be shown in the next section, which now can be expressed as:

$$Q_{SA} = MHSA(Q_T, Q_T) \tag{5}$$

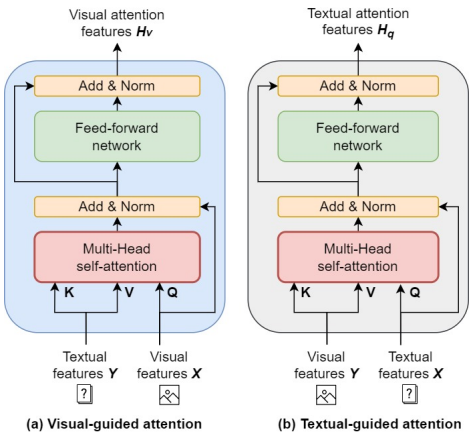


Fig. 2: Two types of Guided Attention modules which are used in our models: (a): Visual-guided attention: to learn the visual features guided by question; (b) Textual-guided attention: to learn the textual attention features guided by image.

3.3 Multi-branch contextual attention

Attention in VQA is motivated by how we pay attention to different regions of an image if given the words of a question, and vice versa. Thus, we demonstrate a multi-branch contextual attention fusion to hierarchically integrate contextual features of image and question. Firstly, we introduce a guided self-attention module to learn the attention of global and local contextual visual features guided by the question, then we show how we fuse different contextual features into a joint features representation to predict the best answer.

Guided attention. The guided attention is inspired by the attention mechanism of Transformer [1]. Given the input features X and Y, the goal of guided attention is to capture the intra-interaction amongst all entities in X guided by Y. Firstly, the attention sub-layer projects the input into query matrix Q, key matrix K, and value matrix V embeddings by three linear projections:

$$Q = X \cdot W^q \qquad K = Y \cdot W^k \qquad V = Y \cdot W^v \tag{6}$$

where $W^q \in \mathbb{R}^{d_x \times d}$, $W^k \in \mathbb{R}^{d_y \times d}$, $W^q \in \mathbb{R}^{d_y \times d}$, are learnable weight matrices; d_x, d_y are dimensions of input X, Y, and d is the common hidden dimension.

Then, by multiplication between each feature of Q and K, we obtain the attention weights matrix for aggregating information from X and Y. Following by normalizing the attention weights with the square root of the hidden dimension d and applying a softmax non-linearity function, we obtain the guided attention

weight A by weighted summation over all values V:

$$A = ATT(X, Y) = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{7}$$

To get better feature representation, we usually use a multi-head mechanism. Each head is an independent attention operation that runs multiple times in parallel. The independent attention outputs are simply concatenated and linearly transformed into the original dimensions.

$$\begin{aligned} F = MHSA(X, Y) &= concat(head_1, ..., head_H).W^o \\ head_h = ATT_h(X, Y) &= softmax\left(\frac{Q_h K_h^T}{\sqrt{d_h}}\right)V_h \\ &= softmax\left(\frac{(XW_h^q)(YW_h^k)^T}{\sqrt{d_h}}\right)(YW_h^v) \end{aligned} \tag{8}$$

where $W_h^q \in \mathbb{R}^{d_x \times d_h}$, $W^k \in \mathbb{R}^{d_y \times d_h}$, $W^q \in \mathbb{R}^{d_y \times d_h}$, are the projection matrices of $head_h$; $W^o \in \mathbb{R}^{H*d_h \times d}$ is the projection matrix for all H heads. d_h is the dimensionality of the output features from each head. To avoid the overgrowth of computational cost, d_h is usually set to $d_h = d/H$.

After computing the guided attention features using multi-head attention, skip-connection followed by a layer normalization $LayerNorm(\cdot)$ [35]. is applied to enhance optimization and training stabilization:

$$F' = LayerNorm(MHSA(X, Y) + X) \tag{9}$$

Finally, the guided attention features F' are passed through a pointwise feed-forward network (FFN), and another layer normalization with skip-connection is also applied to get the ultimate output H, similar to the Transformer encoder:

$$H = LayerNorm(FFN(F') + F') \tag{10}$$

The $FFN(\cdot)$ consists of two fully–connected layers, each one followed by a ReLU activation function and dropout.

$$FFN(X) = ReLU\{[ReLU\,(XW_1 + b_1)]\,W_2 + b_2\} \tag{11}$$

where $W_1 \in \mathbb{R}^{d_{in} \times d_m}, W_2 \in \mathbb{R}^{d_m \times d_o}, b_1 \in \mathbb{R}^{d_m}, b_2 \in \mathbb{R}^{d_o}$ are the parameters, with d_{in}, d_m, d_o are the input, hidden and output dimensions of FFN, respectively.

The final output H should have the same dimensions as the input X. Generally, all above procedures of Guided-attention (GA) module can be formulated:

$$H = GA(X, Y) \tag{12}$$

Multi-context features fusion. The guided attention module will apply to both global and local contextual visual features, corresponding to the question, in two different branches. The visual-guided attention will take the input image

features as X, question features as Y, and generate the attended global contextual features $H_G \in \mathbb{R}^{N \times n_L}$ and local contextual features $H_L \in \mathbb{R}^{M \times n_G}$. In other words, the output attended visual features will represent the focused regions of the image related to the information in the question.

$$H_L = GA(V_L, Q_{SA}) \tag{13}$$

$$H_G = GA(V_G, Q_{SA}) \tag{14}$$

The output attended visual features H_L and H_G already contain rich information of the image. However, to refine the significant features from the contextual information and fuse them into a multi-context feature representation, an attentional reduction is applied. Attentional reduction is indeed a top-down attention mechanism [8]. It contains the aforementioned FFN layer with hidden and output dimensions of $d_{in}/2$ and 1, a softmax function to generate the attentional distribution, then we take the weighted sum over the visual features. Taking H_L as an example of attended visual features, we can formulate as follows:

$$H'_L = \sum_{i=1}^{M} \alpha_i . h_L^{(i)} \qquad \text{with } \alpha_L = softmax(FFN(H_L)), \tag{15}$$

where $\alpha_L = [\alpha_1, ..., \alpha_M]$ the learned attention weights.

With the computed attended global and local features of the image, we fuse them into a multi-vision embedding vector by a linear combination:

$$V_{GL} = H'_G \oplus H'_L \tag{16}$$

The linear combination can be element-wise multiplication, summation, or concatenation. However, we found that the concatenation, which creates a two-glimpse attended visual features vector, can yield better performance.

Visual-and-textual features representation. The multi-vision embedding V_{GL} and the question features Q_{SA} again go through a textual-guided attention module. This textual guided attention is similar to visual guided attention, however, this time, the input X is the textual features and Y is the visual features, thus generates the textual attention features Q_{GA}.

$$Q_{GA} = GA(Q_{SA}, V_{GL}) \tag{17}$$

This textual-guided attention enhances the question features by focusing on the important words in the question guided by the image.

At this point, we already have two significant visual attended and textual attended features. We need to perform feature fusion to obtain a visual-and-textual features representation J. The visual and textual features vectors are passed through linear projection layers then combined by a Hadamard product. The calculation can be expressed as follow:

$$J = LayerNorm\left(W_a Q_{GA} \odot W_b V_{GL}\right) \tag{18}$$

where $W_a \in \mathbb{R}^{1024 \times d_a}, W_b \in \mathbb{R}^{1024 \times d_b}$ are the projection matrices, layer normalization is also used to stabilize training.

3.4 Answer Prediction

In the manner of most VQA research, we consider the VQA task as a classification task, the final joint features representation J will be fed into the classifier layer consisting of two fully connected networks, which is actually FFN, then applied a softmax function for final prediction:

$$\hat{p} = softmax(FFN(J)) \tag{19}$$

where $FFN(\cdot)$ used here has the hidden and output dimension of 512 and $|\mathcal{A}|$, which is the number of candidate answers $|\mathcal{A}| = 353$.

3.5 Loss function

We optimize our model with the cross-entropy loss (CE) as a loss function for training, which is calculated as

$$L_{CE} = -\sum_{i=1}^{|\mathcal{A}|} y_i.log(\hat{p}_i) \tag{20}$$

with y_i is the ground-truth answer, $\hat{p}_i$ is the predicted probability of i-th class.

4 Experiment

4.1 Dataset and Evaluation metric

Dataset. We experiment our model on a Vietnamese VQA dataset, ViVQA [11]. The images in ViVQA are obtained from the MS-COCO[1] source, the question-answer (Q-A) pairs are translated into Vietnamese by machine translation, and have Vietnamese native speakers tune the sentences to avoid ambiguity. Therefore, the dataset is more accurate and natural than being automatically translated from English, making ViVQA a good-quality one to perform benchmarks in the VQA task for Vietnamese. ViVQA contains 11,999 Q-A pairs in the train set and 3,001 Q-A pairs in the test set with a total of 10,328 images.

Evaluation metric. By considering the VQA task as classification, we evaluate our model on ViVQA using *accuracy* as the metric.

4.2 Experiment settings

Model hyper-parameters. We conduct the experiments to evaluate our model with hyper-parameters as follows. The size of the input image I is reshaped to $224 \times 224 \times 3$, then we extract the global visual features from hidden states of pre-trained ViT-Base's last layer with dimension 197×768, and local visual

[1] https://cocodataset.org/

Table 1: Detail value of training hyper-parameters

Hyper-parameter	**Value**
No. of epochs	40
Batch size	32
Initial learning rate (LR)	$1e-4$
LR scheduler	linear decay with 5 warm-up epochs
Adam's betas (β_1, β_2)	$(0.9, 0.999)$
Weight decay λ	$1e-5$
Dropout ratio	0.3

Table 2: Comparisons of our model with former baseline methods

Models	**Accuracy (%)**
Hierarchical Co-Att + PhoW2Vec	34.96
LTSM + PhoW2Vec	33.85
Bi-LTSM + PhoW2Vec	33.97
Our model	**60.76**

features from output of Conv5 layer of pre-trained ResNet-34 with features map of $7 \times 7 \times 512$. For the input question, we truncate the question to a maximum length of 20 and do padding with zero vector when its length is less than 20. The extracted question features are obtained from the last hidden states of phoBERT with a dimension of 20×768. In all guided attention modules, we set the number of heads $H = 8$, so the latent dimensionality of each head is $d_h = d/H = 768/8 = 96$; the input, hidden and output dimensions of FFN is $768, 1536, 768$, respectively.

Training hyper-parameters. We trained the model on a machine with a Tesla P100-PCIE-16GB GPU. We use Adamax [36] as the optimizer to update parameters, the other training hyper-parameters are defined detail in Table 1.

4.3 Comparison with other methods

As there are few published works on VQA tasks in the Vietnamese. Therefore, we evaluate and compare our model with 3 baseline methods proposed by the authors in ViVQA paper [11]. All three models use ResNet for visual representations and PhoW2Vec for textual features. Each one has a different method for joint features understanding: the first one uses Long Short Term Memory (LSTM) for associative attribute learning between images and questions, the second one enhances by a Bidirectional Long Short Term Memory (Bi-LSTM), and the last one used the Hierarchical Co-Attention [13] method to learn co-attention

Table 3: First experiment result on ViVQA test set

Visual extractors	**Accuracy (%)**
ResNet-34	58.09
ViT-base	59.51
ViT + ResNet	**60.76**

Table 4: Second experiment result on ViVQA test set

Visual features combination	**Accuracy (%)**
Summation	58.91
Element-wise product	59.50
Concatenation	**60.76**

of both features. Details of the result are shown in Table 3. Undoubtedly, our model outperforms with a much more significant result compared to the former baseline. Our model achieve the accuracy of 60.76% on the test set, which is approximately 1.5 times better than the best baseline method. This result demonstrates the effectiveness of our proposed method compared to others.

4.4 Ablation study

To explore the effectiveness of our Multi-vision Embeddings for Contextual Attention learning, we conduct two sets of experiments to study: (1) the better efficiency of extracting both global and local visual features over the single representation, (2) the best linear combination of two guided-attention visual features.

In the first experiment, we train two more models, one using ResNet-34 only and the other one using ViT-base only for image understanding. We compare the results between single visual features and multi-vision contextual features. The evaluation results on the test set are shown in Table 3. We conclude that the combination of both global and local contextual visual features improves the performance and yields a better result than a single representation.

In the second experiment, we train our model with three types of linear combinations of contextual visual features: concatenation, summation, and element-wise product. Results are shown in Table 4, indicating that concatenation is the best choice to combine visual features into a multi-vision contextual representation.

5 Conclusion

In this paper, we proposed a Multi-vision Embedding based on the hypothesis that both global and local contextual features of the image can enhance the visual understanding for VQA task. In addition, the textual features is obtained by phoBERT, a BERT variant for Vietnamese, to learn the attentional embedding of the input question. Based on the extracted visual and textual features, we designed a multi-branch contextual attention fusion that hierarchically integrates visual and textual attention features that guided by each other and fuse them into a multimodal features representation. We also performed extensive experiments on ViVQA, a Vietnamese VQA dataset, to demonstrate the effectiveness and outstanding performance of our model compared to former baselines with the resulting accuracy of 60.67%.

Acknowledgments

This research is supported by research funding from Faculty of Information Technology, University of Science, Vietnam National University - Ho Chi Minh City.

References

[1] Ashish Vaswani et al. "Attention is All you Need". In: *Advances in Neural Information Processing Systems*. Vol. 30. Curran Associates, Inc., 2017.

[2] Mingxing Tan and Quoc Le. "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks". In: *Proceedings of the 36th International Conference on Machine Learning*. PMLR, May 24, 2019, pp. 6105–6114.

[3] Yanwei Li et al. "Voxel Field Fusion for 3D Object Detection". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 1120–1129.

[4] Shaolei Zhang and Yang Feng. "Gaussian Multi-head Attention for Simultaneous Machine Translation". In: *arXiv preprint arXiv:2203.09072* (2022).

[5] Xiujun Li et al. "Oscar: Object-Semantics Aligned Pre-training for Vision-Language Tasks". In: *Computer Vision – ECCV 2020*. Ed. by Andrea Vedaldi et al. Lecture Notes in Computer Science. Cham: Springer International Publishing, 2020, pp. 121–137. ISBN: 978-3-030-58577-8. DOI: 10.1007/978-3-030-58577-8_8.

[6] Tung Le, Huy Tien Nguyen, and Minh Le Nguyen. "Multi visual and textual embedding on visual question answering for blind people". In: *Neurocomputing* 465 (Nov. 20, 2021), pp. 451–464. ISSN: 0925-2312. DOI: 10.1016/j.neucom.2021.08.117.

[7] Rowan Zellers et al. "From Recognition to Cognition: Visual Commonsense Reasoning". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2019, pp. 6720–6731.

[8] Peter Anderson et al. "Bottom-Up and Top-Down Attention for Image Captioning and Visual Question Answering". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2018, pp. 6077–6086.

[9] Zhou Yu et al. "Deep Modular Co-Attention Networks for Visual Question Answering". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2019, pp. 6281–6290.

[10] Jiasen Lu et al. "ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks". In: *Advances in Neural Information Processing Systems*. Vol. 32. Curran Associates, Inc., 2019.

[11] K. Tran et al. "ViVQA: Vietnamese Visual Question Answering". In: *PACLIC*. 2021.

[12] Stanislaw Antol et al. "VQA: Visual Question Answering". In: *Proceedings of the IEEE International Conference on Computer Vision*. 2015, pp. 2425–2433.

[13] Jiasen Lu et al. "Hierarchical Question-Image Co-Attention for Visual Question Answering". In: *Advances in Neural Information Processing Systems*. Vol. 29. Curran Associates, Inc., 2016.

[14] Zhou Yu et al. "Multi-Modal Factorized Bilinear Pooling With Co-Attention Learning for Visual Question Answering". In: *Proceedings of the IEEE International Conference on Computer Vision*. 2017, pp. 1821–1830.

[15] Shaoqing Ren et al. "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks". In: *Advances in Neural Information Processing Systems*. Vol. 28. Curran Associates, Inc., 2015.

[16] Alexey Dosovitskiy et al. "An image is worth 16x16 words: Transformers for image recognition at scale". In: *arXiv preprint arXiv:2010.11929* (2020).

[17] Maithra Raghu et al. "Do Vision Transformers See Like Convolutional Neural Networks?" In: *Advances in Neural Information Processing Systems*. May 21, 2021.

[18] Sepp Hochreiter and Jürgen Schmidhuber. "Long Short-Term Memory". In: *Neural Computation* 9.8 (Nov. 1997). Pp. 1735–1780. ISSN: 0899-7667. DOI: 10.1162/neco.1997.9.8.1735.

[19] Jeffrey Pennington, Richard Socher, and Christopher D. Manning. "Glove: Global vectors for word representation". In: *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*. 2014, pp. 1532–1543.

[20] Jacob Devlin et al. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. arXiv, May 24, 2019. arXiv: 1810.04805[cs].

[21] Karen Simonyan and Andrew Zisserman. "Very deep convolutional networks for large-scale image recognition". In: *arXiv preprint arXiv:1409.1556* (2014).

[22] Kaiming He et al. "Deep Residual Learning for Image Recognition". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2016, pp. 770–778.

[23] Moshiur R. Farazi and Salman H. Khan. "Reciprocal attention fusion for visual question answering". In: *arXiv preprint arXiv:1805.04247* (2018).

[24] Hangbo Bao, Li Dong, and Furu Wei. "Beit: Bert pre-training of image transformers". In: *arXiv preprint arXiv:2106.08254* (2021).

[25] Ze Liu et al. "Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2021, pp. 10012–10022.

[26] Stéphane D'Ascoli et al. "ConViT: Improving Vision Transformers with Soft Convolutional Inductive Biases". In: *Proceedings of the 38th International Conference on Machine Learning*. PMLR, July 1, 2021, pp. 2286–2296.

[27] Davide Alessandro Coccomini et al. "Combining efficientnet and vision transformers for video deepfake detection". In: *International Conference on Image Analysis and Processing*. Springer, 2022, pp. 219–229.

[28] Feng Yan, Wushouer Silamu, and Yanbing Li. "Deep Modular Bilinear Attention Network for Visual Question Answering". In: *Sensors (Basel, Switzerland)* 22.3 (Jan. 28, 2022), p. 1045. ISSN: 1424-8220. DOI: 10.3390/s22031045.

[29] Ranjay Krishna et al. "Visual Genome: Connecting Language and Vision Using Crowdsourced Dense Image Annotations". In: *International Journal of Computer Vision* 123.1 (May 1, 2017), pp. 32–73. ISSN: 1573-1405. DOI: 10.1007/s11263-016-0981-7.

[30] Andreas Chandra, Affandy Fahrizain, and Simon Willyanto Laufried. "A survey on non-english question answering dataset". In: *arXiv preprint arXiv:2112.13634* (2021).

[31] Kiet Nguyen et al. "A Vietnamese Dataset for Evaluating Machine Reading Comprehension". In: *Proceedings of the 28th International Conference on Computational Linguistics*. COLING 2020. Dec. 2020, pp. 2595–2605. DOI: 10.18653/v1/2020.coling-main.233.

[32] Kiet Van Nguyen et al. "Enhancing Lexical-Based Approach With External Knowledge for Vietnamese Multiple-Choice Machine Reading Comprehension". In: *IEEE Access* 8 (2020). Pp. 201404–201417. ISSN: 2169-3536. DOI: 10.1109/ACCESS.2020.3035701.

[33] Shikhar Tuli et al. "Are Convolutional Neural Networks or Transformers more like human vision?" In: *arXiv preprint arXiv:2105.07197* (2021).

[34] Dat Quoc Nguyen and Anh Tuan Nguyen. "PhoBERT: Pre-trained language models for Vietnamese". In: *arXiv preprint arXiv:2003.00744* (2020).

[35] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. "Layer normalization". In: *arXiv preprint arXiv:1607.06450* (2016).

[36] Diederik P. Kingma and Jimmy Ba. "Adam: A method for stochastic optimization". In: *arXiv preprint arXiv:1412.6980* (2014).